

Script

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1 Introduction

Pharmaceutical products in US can be broadly classified into two categories:

1. Branded drugs, which are pioneer molecules which enjoy market exclusivity for a limited period, usually through patents.
2. Generic drugs that are bioequivalent to their branded counterparts.

After loss of market exclusivity - usually through patent expiration - generics enter the market and compete with each other and the branded product.

read off slides until the "dynamics" slide. Some helpful lines

1. Use this in the Entry Delays slides:
 - To market their products, a generic manufacturer first needs to gain FDA approval.

Generic manufacturers have to prove that they can manufacture the drug accurately and at scale, and according to Good Manufacturing Practices.

Approval times of generics is very lengthy - with a mean approval time around 40 months - and highly stochastic, as the amount of work required for verification varies on a case-by-case basis.

2. Use this in the Authorized Generics slides:

“On the other hand, the AG does not need FDA approval to launch, and so can be released any time.”

1.1 Dynamics slides

How does the market evolve after loss-of-exclusivity?

General pricing patterns are:

- Brand drug charges high price.
- Generic and AG charge low prices.
- As more generics enter, prices of nonbrands fall, but brand price stays high.

Here is a time series of market share and prices for a representative molform.

- LOE happens at $t = 0$.
- (point to right figure) Over time, brand price stays high, while AG and generics priced low.
- (point to left figure) Brand loses mkt share, generic and AG gains market share.

Read off Research question and Contributions slide

2 Data

Read off data and descriptive statistics slides

Emphasize: We restrict our attention to Prescription drugs.

2.1 Generic entry rates slides

Next we look at the generic entry rate in our dataset.

We define the “generic entry rate” at a time-since-LOE as the number of generic entries in that time divided by the total number of generic entries.

Figure on left shows the generic entry rate by quarters since loss-of-exclusivity. 33% of entries happen in the quarter of LOE, 8% in the first quarter after LOE, 11% in the second quarter after LOE, and so on.

Figure on right is the cumulative version of the left figure.

(point to left figure) The most notable pattern is that a fraction of the generic entry happens in the quarter of LOE (around 33% of all generic entries). This can be explained by 2 factors. First, many generics apply for FDA approval well in advance of LOE, and a fraction get approval before the LOE period. Second, we define LOE to be the quarter when the first generic enters, so that mechanically drives some of the pattern.

(point to right figure) However, the majority of generic entry happens **after** the first quarter. By quarter 20 after LOE, about 93% of the generic entries have taken place.

2.2 AG release timing slides

Figure shows when the AGs were released in these markets relative to generic entry.

Emphasize: Recall that AGs can be released at any time, and do not need additional FDA approval to launch.

This is a histogram that shows how many quarters after first-generic-entry is AG released.

- Of the 109 molforms that see AG released, 58 markets see AG release immediately with the first generic entry, 10 markets see AG release one quarter after the first generic entry, and 6 markets see AG released two quarters later.

3 Structural model

We model demand in two stages. In the first stage, for a molecule-formulation the consumer chooses whether to buy a branded product, a nonbranded product, or go for the outside option (no drug or a different molecule). **(Emphasize) Think of this as the physician and patient deciding whether to go for a brand or nonbrand version of a molform.**

The utility the consumer gets (read from slides upto eqn) is a function of: molform fixed effect, quarter fixed effect, log of average price in the group, brand group indicator, brand-specific time trend, unobserved group -specific demand shock, and logit error. We have random coefficients on the price and brand which are distributed normally.

The brand group and outside option group have a single product each, so if those are chosen, there is no choice to be made in the second stage. But if the consumer chooses a nonbranded product, then in the second stage, the consumer chooses which nonbranded product to buy. **(Emphasize) Think of this as: once the patient decides to go for a nonbrand group, the pharmacy picks the product.**

The utility of (read from slides upto equation) is a function of: molform fixed effect, log of price of the nonbrand product, Authorized Generic indicator, unobserved product-specific demand shock, and logit error.

Given our assumptions, the market share of product j is probability of the consumer choosing group g times the probability of the consumer choosing product j conditional on group g .

(Emphasize) Generally, we regard Stage 1 as representing a decision made by the patient, physician, and insurer combined; while Stage 2 represents a decision made by pharmacies, procurement officers at providers (e.g. hospitals, clinics, etc.), and other sellers.

So to recap the demand model: the first stage is consumer choosing whether to buy a branded product, a nonbranded product, or go for the outside option (no drug or a different molecule). The second stage is the consumer choosing which nonbranded product to buy, if

they chose the nonbranded group in the first stage.

3.1 Overview of supply side model/timeline

The supply side is a model with 2 stages:

Read off of slides.

(After reading the "first stage", say) Generic entry is modeled as a static entry game because generics cannot exactly choose when to launch; they have to wait for FDA approval, which is unpredictable.

3.2 Second stage

3.2.1 Per period profit slides

Brand drug maker's per-period profit is: (price of brand drug - marginal cost of brand drug) times brand drug market share times market size + if there is an AG, then (price of AG - marginal cost of AG) times AG market share times market size.

Generic manufacturer's per-period profit is: (price of generic - marginal cost of generic) times generic market share times market size.

I mentioned prices in the previous slide, so how are we predicting prices in my structural estimation?

We do something simple: we regress prices on observed state variables, and use the regression to predict prices in different states.

Why do we do this? Not clear that a single pricing conduct assumption such as Nash-Bertrand really captures the heterogeneity in price setting that happens - think of PBMs, Medicare, hospitals, etc. all bargaining differently with pharma manufacturers.

This method lets us avoid taking a stand on that, and so avoids misspecification issues.

(next slide)

n_{em} is the total number of generics that apply for entry in molform m . This will be

determined in Stage 1, which I will explain a little later. We assume that n -em is known to everyone in the second stage.

At $t = 2$, loss of exclusivity happens. You can think of this as the patent of the branded drug expiring.

- From then on, of all the generics which applied for FDA approval, every period a random number of them get FDA approval and launch.
- Also every period, the brand drug manufacturer decides if it wants to release the AG or stay out. Releasing the AG is an irreversible decision.

We set the terminal period T to be 32 quarters after loss of exclusivity.

3.2.2 Value function slides

Since AG release is a dynamic decision, the brand drug maker's choice problem can be written in terms of value functions. Thus, the value function for a brand drug maker every period is given by:

Value function in STATE s - t with logit draw vector $\epsilon_{s,t}$ = profit of brand drug maker - AG entry cost if AG released this period + continuation value + choice-specific logit shock.

This value function reflects the fact that releasing the AG is a timing problem, and so is a dynamic decision. If it releases an AG now, it incurs the entry cost now, but gets a stream of payoffs from AG sales in the future.

The value function for the generic maker is its profit plus the continuation value. Generics do not have to make a dynamic decision in the second stage. However, they will use the value function to make entry decision in the first stage.

After terminal period cap T , profits are zero.

- This accounts for small economic profits in **mature drug markets** due to new drug discovery, patent expiration of other brand drugs, and the extremely-low profit margins

generally observed in a mature generic drug market

3.2.3 Binomial approval rate slide

I've been speaking about the approval rate of generics by the FDA, so how does that feature in the model?

- We model the FDA approval rate as a binomial process.
- (from slides) Suppose by period t in market m , N_{-mt} generics have already received approval.
- (point to λ on slides) λ of t is the probability of a single generic getting approved t periods after LOE; therefore, λ of it is the FDA's approval rate.
- The probability that k of $n - N$ unentered generics get approved in period t is: $n - N$ choose k , times λ raised to k which is the number of generics getting approval, and $1 - \lambda$ raised to $n - N - k$, which is the number of generics not getting approval.
- in our counterfactual, we will increase λ to simulate faster FDA approval rates.

3.3 First stage

Read off slides until generic entry condition equation.

κ_{gm} is the generic entry cost for molecule m .

At the equilibrium number of generic entrants n_{em} , the discounted sum of profits to a single generic weakly covers the entry cost; but if an additional generic were to enter, it will no longer cover the entry cost.

In other words, generics enter until their ex-ante value function weakly covers their entry cost.

4 Estimation

Read off slides until marginal cost part.

Marginal cost:

- Recall that we are predicting prices using a flexible regression and not by assuming a specific pricing conduct. Therefore, we cannot directly invert out the MCs from observed prices.
- Instead we estimate MC to rationalize observed generic numbers.
- (ϑ not ν)

We set $MC_m = \vartheta \bar{p}_{gm}$, where $\bar{p}_{gm}$ is the avg generic prices in m

- We estimate ϑ to match observed generic numbers.
- How does this work? The intuition is that higher ϑ will scale down predicted profits so that at the publicly-reported generic entry costs, the model predicts reasonable generic entry numbers

The AG's logit scaling parameter is estimated via maximum likelihood.

We calibrate AG entry cost to zero. This is because, the majority of a generic's sunk cost of entry mostly comes from having to reverse-engineer the production process of the brand drug and setting up the supply chains; this is not a cost the AG release has to incur.

Our demand estimation results are as expected: demand declines with price, is higher for brand and AGs, and declines for the brand over time. We estimate the variance of the random coefficients to be 0.

Our supply estimation results show that ϑ lies between ... (read off slides).

5 Counterfactuals

Now that we have our model and estimates, we can use them to conduct counterfactuals.

We simulate what happens to market outcomes if generic entry delays are reduced, and if AGs are banned.

We do so by fully solving for AG policy function and n-em, running simulations, and reporting average outcomes.

Before diving into counterfactuals, I will briefly discuss profitability in different phases of a molecule-formulation's post-LOE lifecycle:

- The early quarters of a molecule-formulation after loss-of-exclusivity are the most profitable for firms.
- During these periods, there are few firms in the market, resulting in high markups and high economic profits.
- As time goes by, more generics enter the market, resulting in markups being driven close to zero and low economic profits.

5.1 Reduced entry delays

In the first counterfactual we focus on reduced entry delays, which translates to faster generic approval rates in our model.

Faster approval rate leads to two opposing forces on a generic's entry decision.

- Faster approval means a generic gets to launch earlier, and so has more time to market its product and make profits. Therefore, this higher profit potential leads to greater incentive to enter.
- On the other hand, faster approval rate also means that rivals can launch earlier too. This means that by the time a generic launches, more of its rivals will be present in the market. This leads to more intense price competition and lower profits, leading to smaller incentive for entry.

In simulations, (2) dominates... (read off slides till the end).

For AG, the first mechanism isn't present, since an AG can choose to enter whenever it wants. However, the second mechanism is present.

5.2 AG ban

Next, we study the short-run and long-run effects of AG presence. To that end, we study market outcomes when allowing AG to be released versus an AG ban.

- **In both the data and model simulations**, the AG is generally released soon after loss-of-exclusivity. This means that if we ban AGs, there will be one fewer competitor in the early stages of the market.
- However, the AG ban incentivizes more generics to enter, since there is one less competitor.

It turns out that the effect on overall prices is mixed:

- Prices are generally higher in the early periods of the market: Generics who enter early get to enjoy higher prices than otherwise, since AG is not present in the market to create additional price competition.
- The effect on prices in later periods of the market is ambiguous: This is because the AG ban can sometimes incentivize multiple generics to enter, leading to more intense price competition later on.

Why can an AG ban incentivize more than one additional generic to enter the market?

- Since the AG is released early, it has a negative impact on every generic and particularly during the most profitable early periods of the market.
- If the AG is banned, not only is there one fewer competitor, but it is also one less competitor in the most profitable periods in the market.

- This results in a significant increase in expected generic profits compared to one fewer independent generic.
- In turn, this can incentivize multiple new generics to enter than if an AG had been present.

Read off Conclusion slides